

[_3D.StoreType \[SimTalk\]](#)

Sets the type of the *Store* designated by <Path>.

Type

Attribute

Syntax

```
<Path>._3D.StoreType:string
```

Assignment Value

You can assign a value of data type *string*.

You can specify "Floorspace", "Rack with round posts", "Rack with square posts", "Floorspace (dynamic)", "Rack with round posts (dynamic)", or "Rack with square posts (dynamic)".

Example

```
Store3._3D.StoreType := "Rack with round posts (dynamic)"  
Store3._3D.Gap := 0.1  
Store3._3D.FloorThickness := 0.15  
Store3._3D.GroundClearance := 0.3
```

See also

[Edit 3D Properties \[dialog\]](#) > [Appearance of the Store](#) > [Type \[drop-down list\]](#) - Store

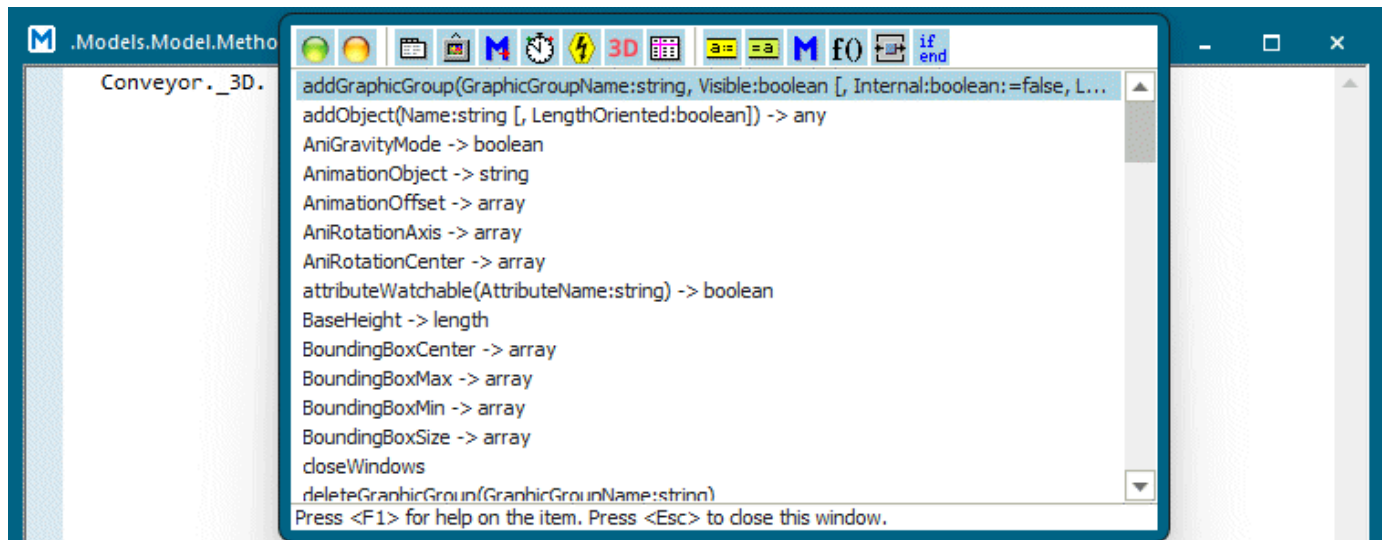
[createRack \[SimTalk\]](#)

Accessing Length-oriented Objects

[__Accessing Length-oriented Objects](#)

SimTalk provides the *attributes* and *methods* listed in the table of contents to the left for defining *length-oriented objects* in 3D.

You can view and access the **attributes** and **methods** for defining *length-oriented objects* by clicking **Auto Complete** on the **Edit** ribbon tab of the *Method Editor*.



See also

[Auto Complete](#)

[_3D.AniGravityMode \[SimTalk\]](#)

Activates **Gravity Mode** of the *length-oriented object* designated by <Path> (`true`) or deactivates it (`false`).

Remarks

This setting only applies to the *length-oriented objects*. It is especially helpful for suspension tracks, etc.

Gravity Mode only affects automatic animations, that is animations for which you did not specify an animation path.

On a suspension track the part is always placed perpendicular to the floor in **Gravity Mode**, either hanging from a rail on the ceiling or standing up on a rail on the floor.

Type

Attribute

Syntax

```
<Path>._3D.AniGravityMode: boolean
```

Assignment Value

You can assign a value of data type *boolean*.

Example

```
MyConveyor._3D.AniGravityMode := true
```

See also

[Gravity Mode](#)

_3D.AnimationOffset [SimTalk]

Sets the **Animation Offset** of the *length-oriented object* designated by <Path>.

Type

Attribute

Syntax

```
<Path>._3D.AnimationOffset:length[3]
```

Assignment Value

You can assign an *array* of data type *length* with three values.

The values set the offset of the animation path to the surface of the object along the **X-direction**, the **Y-direction**, and the **Z-direction**.

Example

```
MyConveyor._3D.AnimationOffset := [0m, 0m, 1m]
```

See also

[Animation Offset](#)

_3D.AniSmoothRotation [SimTalk]

Activates **Smooth Rotation** of the *length-oriented object* designated by <Path> (`true`) or deactivates it (`false`).

Remarks

This setting only applies to the *length-oriented objects*. It is especially helpful for suspension tracks, etc.

Smooth Rotation only affects automatic animations, that is animations for which you did not specify an animation path.

If the object currently animates its *MUs* using a user-defined **Animation Path Type > Lines, Smooth Rotation** specifies whether the part slowly changes its rotation around the anchor points (like this is the case on a *Station* (`true`) for example) or if it abruptly changes its orientation (like at a corner on a *Conveyor* with the standard MU animation) (`false`).

Type

Attribute

Syntax

```
<Path>._3D.AniSmoothRotation:boolean
```

Assignment Value

You can assign a value of data type *boolean*.

Example

```
MyConveyor._3D.AniSmoothRotation := true
```

See also

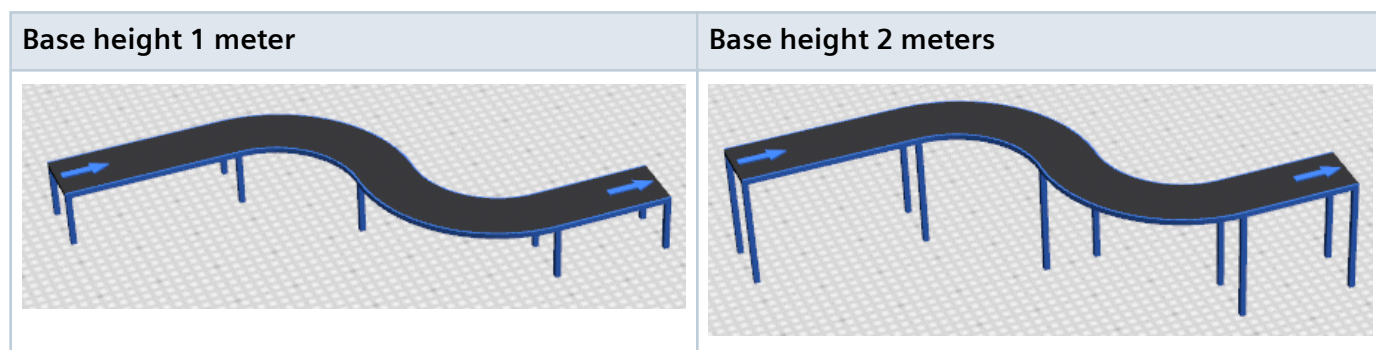
Smooth Rotation

_3D.BaseHeight [SimTalk] - 3D

Sets the **Base Height** of the *length-oriented object* designated by <Path>.

Remarks

The **Base Height** sets the distance between the insertion height (z-coordinate) of the *length-oriented object* and the height of a segment whose ΔZ equals 0.



The **Base Height** can be inherited, meaning that you can define it once in the class instead of having to define it for each instance individually. The z-coordinate defines the floor of the installation and the base height defines the distance of the legs from the floor. By changing the setting ΔZ in the **Segments Table**, you can add an additional offset.

Type

Attribute

Syntax

```
<Path>._3D.BaseHeight:length
```

Assignment Value

You can assign a value of data type *length*.

Example

```
MyConveyor._3D.BaseHeight := 3
print MyConveyor._3D.BaseHeight
```

See also

[Base Height \[tab Appearance\]](#)

[_3D.BaseHeight \[SimTalk\] - 3D](#)

[_3D.getExtSegments \[SimTalk\]](#)

[_3D.setExtSegments \[SimTalk\]](#)

[Insert Curved and Straight Segments](#) > **Anchor Point Height**

_3D.ExtConfiguration [SimTalk]

Sets the extrusion configuration of the *length-oriented object* designated by <Path>.

Remarks

An example of an extrusion configuration might look like this:

.Models.Model.MyDataTable								
	string 0	table 1	integer 2	length[m] 3	length[m] 4	length[m] 5	integer 6	table 7
string		Extrusion Profile	Configuration Type	Start Offset	End Offset	Repetition Interval	Graphic Group ID	Material
1	Legs_1	[2 x 5]	20	-0.09	0	0	0	Material
2	Legs_2	[2 x 5]	20	-0.09	0	0	0	Material
3	Lanes	[2 x 5]	2	0	0	0	0	Material
4	Frame_2	[2 x 5]	0	0	0	0	0	Material
5	Frame_1	[2 x 5]	0	0	0	0	0	Material
6								
7								
8								
9								
10								
11								

.Models.Model.MyDataTable[7,5]						
13395507						
integer 1	integer 2	integer 3	integer 4	real 5	real 6	
string	Diffuse Color	Ambient Color	Specular Color	Emissive Color	Transparency	Shininess
1	13395507	13395507	5855577	0	0	0.267

Type

Attribute

Syntax

```
<Path>._3D.ExtConfiguration:table
```

Parameters of the Extrusion Configuration

The parameters of the extrusion configuration designate the following settings:

- **Index Name:** Is the name of the extrusion group, it is only entered once for each group in the first column of the group, i.e., in column 0.
- **Extrusion Profile:** The subtable [2 x number of points], which you can open by double-clicking in the cell **Extrusion Profile**, specifies the profile that will be extruded according to the settings below. Define the values in a way so that they match a **width** of 1 meter.
- **Configuration Type:** You can set the following values:

Name/Value for Assignment	Description
Path, z max, Move abs/0	Extrude once along the path with z as defined in the anchors, the distance from the edge remains fixed when changing the width.
Path, z max, Move rel/1	Extrude once along the path with z as defined in the anchors and pin the profile base point to its relative position when changing the width.
Path, z max, Scale x/2	Extrude once along the path with z as defined in the anchors and scale in the x-direction when changing the width.
Path, z max, Scale xy/3	Extrude once along the path with z as defined in the anchors and scale in the x-direction and the y-direction when changing the width.
Path, Floor, Move abs/4	Extrude once along the path with z = 0.0, the distance from the edge remains fixed when changing the width.
Path, Floor, Move rel/5	Extrude once along the path with z = 0.0 and pin the profile base point to its relative position when changing the width.
Path, Floor, Scale x/6	Extrude once along the path with z = 0.0 and scale in the x-direction when changing the width.
Path, Floor, Scale xy/7	Extrude once along the path with z = 0.0 and scale in the x-direction and the y-direction when changing the width.
Horizontal, Anchors, z max, Move/8	Extrude horizontally for every anchor point with z as defined in the anchors and pin to the borders when changing the width. The graphic tilts according to the incline.
Horizontal, Anchors, z max, Scale/9	Extrude horizontally for every anchor point with z as defined in the anchors and pin to the center when changing the width. The graphic tilts according to the incline.
Horizontal, Anchors, z 1/5, Move/10	Extrude horizontally for every anchor point with z = 1/5 of the anchor point height and pin to the borders when changing the width.
Horizontal, Anchors, z 1/5, Scale/11	Extrude horizontally for every anchor point with z = 1/5 of the anchor point height and pin to the center when changing the width.
Horizontal, Anchors, Floor, Move/12	Extrude horizontally for every anchor point with z = 0.0 and pin to the borders when changing the width.
Horizontal, Anchors, Floor, Scale/13	Extrude horizontally for every anchor point with z = 0.0 and pin to the center when changing the width.
Horizontal, Interval, z max, Move/14	Extrude horizontally with a fixed interval with z as defined in the anchors and pin to the borders when changing the width. The graphic tilts according to the incline.

Name/Value for Assignment	Description
Horizontal, Interval, z max, Scale/15	Extrude horizontally with a fixed interval with z as defined in the anchors and pin to the center when changing the width. The graphic tilts according to the incline.
Horizontal, Interval, z 1/5, Move/16	Extrude horizontally with a fixed interval with $z = 1/5$ of the anchor point height and pin to the borders when changing the width.
Horizontal, Interval, z 1/5, Scale/17	Extrude horizontally with a fixed interval with $z = 1/5$ of the anchor point height and pin to the center when changing the width.
Horizontal, Interval, Floor, Move/18	Extrude horizontally with a fixed interval with $z = 0.0$ and pin to the borders when changing the width.
Horizontal, Interval, Floor, Scale/19	Extrude horizontally with a fixed interval with $z = 0.0$ and pin to the center when changing the width.
Vertical, Floor, Anchors, Move/20	Extrude vertically for every anchor point from the path anchor to the floor and pin to the borders when changing the width.
Vertical, Floor, Anchors, Fix/21	Extrude vertically for every anchor point from the path anchor to the floor and pin to the center when changing the width.
Vertical, Interval, Move/22	Extrude vertically with a fixed interval from the path anchor to the floor and pin to the borders when changing the width.
Vertical, Interval, Fix/23	Extrude vertically with a fixed interval from the path anchor to the floor and pin to the center when changing the width.
Forward, z max, Anchors, Move abs/24	Repeatedly extrude a small segment in the path direction for each anchor point, the distance from the borders remains fixed when changing the width.
Forward, z max, Anchors, Move rel/25	Repeatedly extrude a small segment in the path direction for each anchor point, and pin the base point of the profile to its relative position when changing the width.
Forward, z max, Anchors, Fix/26	Repeatedly extrude a small segment in the path direction for each anchor point, the distance from the center remains fixed when changing the width.
Forward, z max, Interval, Move abs/27	Repeatedly extrude a small segment in the path direction with a fixed interval, the distance from the borders remains fixed when changing the width.
Forward, z max, Interval, Move rel/28	Repeatedly extrude a small segment in the path direction with a fixed interval, and pin the base point of the profile to its relative position when changing the width.
Forward, z max, Interval, Fix/29	Repeatedly extrude a small segment in the path direction with a fixed interval, the distance from the center remains fixed when changing the width.

Name/Value for Assignment	Description
Vertical, z max, Anchors, Move/30	Extrude vertically for each anchor point at the height of the path anchor point and pin to the borders when changing the width.
Vertical, z max, Anchors, Fix/31	Extrude vertically for each anchor point at the height of the path anchor point and pin to the center when changing the width.
Vertical, z max, Interval, Move/32	Extrude vertically with a fixed interval at the height of the path anchor point and pin to the borders when changing the width.
Vertical, z max, Interval, Fix/33	Extrude vertically with a fixed interval at the height of the path anchor point and pin to the center when changing the width.
Forward, Floor, Anchors, Move abs/34	Repeatedly extrude a small segment in the path direction for each anchor point with $z = 0.0$, the distance from the borders remains fixed when changing the width.
Forward, Floor, Anchors, Move rel/35	Repeatedly extrude a small segment in the path direction for each anchor point with $z = 0.0$, and pin the base point of the profile to its relative position when changing the width.
Forward, Floor, Anchors, Fix/36	Repeatedly extrude a small segment in the path direction for each anchor point with $z = 0.0$, the distance from the center remains fixed when changing the width.
Forward, Floor, Interval, Move abs/37	Repeatedly extrude a small segment in the path direction with a fixed interval with $z = 0.0$, the distance from the borders remains fixed when changing the width.
Forward, Floor, Interval, Move rel/38	Repeatedly extrude a small segment in the path direction with a fixed interval with $z = 0.0$, and pin the base point of the profile to its relative position when changing the width.
Forward, Floor, Interval, Fix/39	Repeatedly extrude a small segment in the path direction with a fixed interval with $z = 0.0$, the distance from the center remains fixed when changing the width.
Vertical, Ceiling, Anchors, Move/40	Extrude vertically for each anchor point from the path anchor point to the ceiling and pin to the borders when changing the width. The end offset denotes the ceiling height.
Vertical, Ceiling, Anchors, Fix/41	Extrude vertically for each anchor point from the path anchor point to the ceiling and pin to the center when changing the width. The end offset denotes the ceiling height.
Vertical, Ceiling, Interval, Move/42	Extrude vertically with a fixed interval from the path anchor point to the ceiling and pin to the borders when changing the width. The end offset denotes the ceiling height.

Name/Value for Assignment	Description
Vertical, Ceiling, Interval, Fix/43	Extrude vertically with a fixed interval from the path anchor point to the ceiling and pin to the center when changing the width. The end offset denotes the ceiling height.
Horizontal, Anchors, z max Gr, Move/44	Extrude horizontally for each anchor point with z as defined in the anchor points and pin to the borders when changing the width.
Horizontal, Anchors, z max Gr, Scale/45	Extrude horizontally for each anchor point with z as defined in the anchor points and pin to the center when changing the width.
Horizontal, Interval, z max Gr, Move/46	Extrude horizontally with a fixed interval with z, as defined in the anchor points and pin to the borders when changing the width.
Horizontal, Interval, z max Gr, Scale/47	Extrude horizontally with a fixed interval with z, as defined in the anchor points and pin to the center when changing the width.

- **Start Offset:** Is the offset of the start reference in the extrusion direction.
- **End Offset:** Is the offset of the end reference in the extrusion direction.
- **Repetition Interval:** For the configuration types 14, 15, 16, 17, 18, 19, 22, and 23 the value 0 means that the extrusion is being generated once at the start point of the object. A value greater than 0 means that the extrusion is generated every number of meters that you enter. If you specify 0.5, the extrusion will be generated every 50 cm along the <Path>. For the other values, this value is meaningless, and *Plant Simulation* only accepts 0.
- **Graphic Group ID:** By default this value is 0. A value between 0 and 10 generates additional graphic groups into which the program places these thus marked extrusions. The *Fluid object Pipe* for example uses this feature.
- **Material:** In the subtable **Material**, which you can open by double-clicking in the cell **Material**, you can set the following:

The **Diffuse Color [MUs]**, compare **makeRGBValue [SimTalk]**.

The **Ambient Color [MUs]**, compare **makeRGBValue [SimTalk]**.

The **Specular Color [MUs]**, compare **makeRGBValue [SimTalk]**.

The **Emissive Color [MUs]**, compare **makeRGBValue [SimTalk]**.

The **Transparency [MUs]**.

The **Shininess [MUs]**.

Assignment Value

You can a value of data type *table* containing the extrusion configuration.

Note

This attribute returns a copy of the values.

Compare these examples:

```
Conveyor._3D.ExtConfiguration := MyExtrusionConfigurationTable    //  
assigns a new value  
MyExtrusionConfigurationTable := Conveyor._3D.ExtConfiguration    //  
reads a value
```

This example seems to have the correct syntax, but does not have any effects:

```
Conveyor._3D.ExtConfiguration[2,1] := 2  
// does not assign the subtable to the first cell of the attribute
```

To get the desired result of the above instruction, enter this source code:

```
var extConfTable := Conveyor._3D.ExtConfiguration  
extConfTable[2,1] := 2  
Conveyor._3D.ExtConfiguration := extConfTable
```

Example

```
Conveyor._3D.ExtConfiguration := MyExtrusionConfigurationTable  
MyExtrusionConfigurationTable := Conveyor._3D.ExtConfiguration
```

Note

To copy the attribute *_3D.ExtConfiguration* to a *DataTable* in a *Frame*, this does not work via a simple assignment with `:=` but by copying format and contents of the *DataTable*.

Compare the example below:

```
var MyExtrusionConfigurationTable : table := Conveyor._3D.ExtConfiguration  
MyDataTable.delete  
MyExtrusionConfigurationTable.copyFormatTo(MyDataTable)  
MyExtrusionConfigurationTable.copyRangeTo({0,0}..{*,*}, MyDataTable, 0, 0)
```

See also

[Appearance of the Length-oriented Objects](#) in the dialog **Edit 3D Properties**

[User-defined \[Conveyor type\]](#)

makeRGBValue [SimTalk]

[_3D.getExtSegments \[SimTalk\]](#)

Gets the segments table of the *length-oriented object* designated by <Path>.

Type

Method

Syntax

```
<Path>._3D.getExtSegments(Table:table)
```

Parameter

The parameter *Table* of data type *table* designates the table into which *Plant Simulation* writes the settings of the segments table.

Example

```
MyConveyor._3D.getExtSegments(extSegmentsTable)
```

See also

_3D.setExtSegments [SimTalk]	Length [SimTalk] - animatable object
Segments [tab Appearance]	Reuse the Segments of a Length-oriented Object

[_3D.setExtSegments \[SimTalk\]](#)

Sets the segments table of the *length-oriented object* designated by <Path>.

Type

Method

Syntax

```
<Path>._3D.setExtSegments(Table:table)
```

Parameter

The parameter *Table* of data type *table* designates the table from which *Plant Simulation* imports the settings of the segments table.

Example

```
MyConveyor._3D.setExtSegments(extSegmentsTable)
```

See also

_3D.getExternalSegments [SimTalk]	Length [SimTalk] - animatable object
Segments [tab Appearance]	Reuse the Segments of a Length-oriented Object

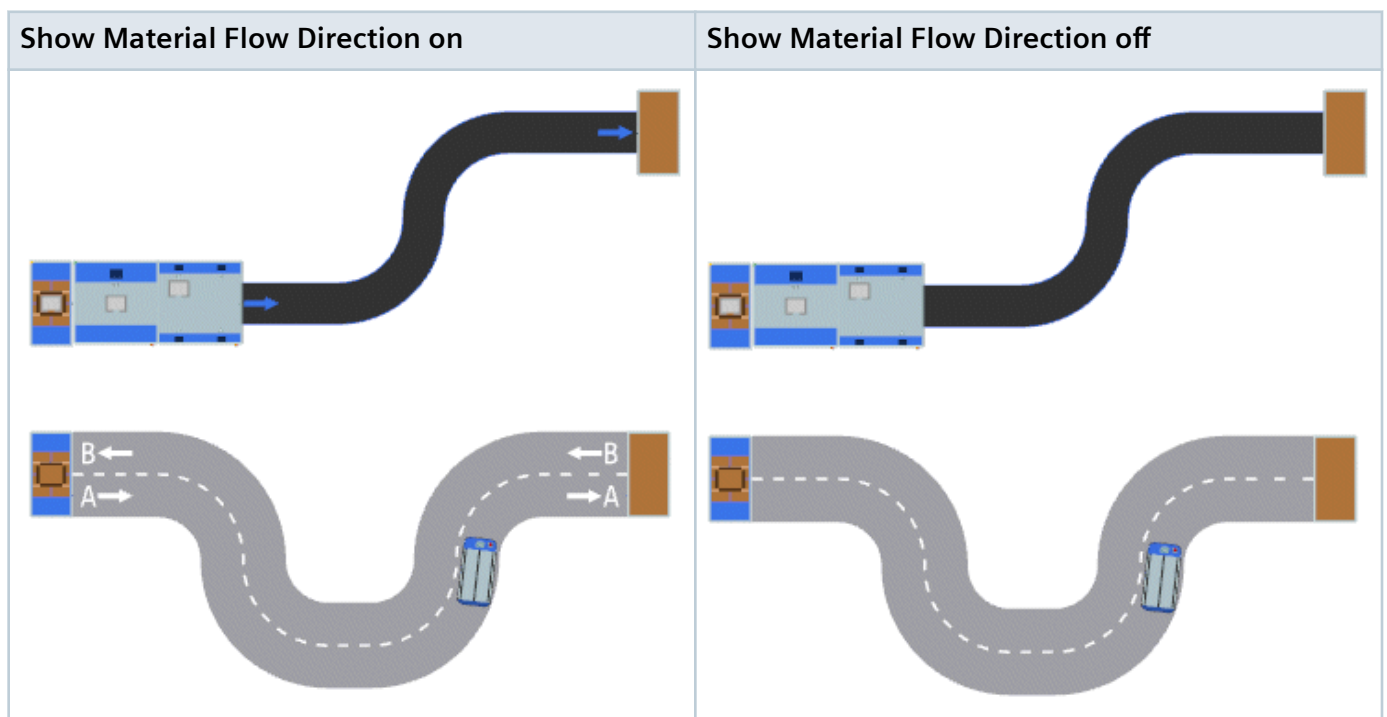
[_3D.ShowMaterialFlowDirection \[SimTalk\]](#)

Shows the material flow direction arrow of the length-oriented object designated by <Path> (*true*) along the insertion direction or hides it (*false*).

Remarks

_3D.ShowMaterialFlowDirection only applies to simulation objects and animatable objects which are length-oriented and which have a meaningful material flow direction.

As the *FootPath* does not have a meaningful material flow direction, it does not show the direction arrow despite being length-oriented.



Type

Attribute

Syntax

```
<Path>._3D.ShowMaterialFlowDirection:boolean
```

Assignment Value

You can assign a value of data type *boolean*.

Example

```
MyConveyor._3D.ShowMaterialFlowDirection := false
```

See also

[Show Material Flow Direction \[check box\]](#)

[_3D.ShowSensors \[SimTalk\]](#)

Shows (`true`) or hides (`false`) the sensors of the *length-oriented object* designated by `<Path>` in 3D.

Type

Attribute

Syntax

```
<Path>._3D.ShowSensors: boolean
```

Assignment Value

You can assign a value of data type *boolean*.

Example

```
MyConveyor._3D.ShowSensors := false
```

See also

[Sensors \[general description\]](#)

[Show Sensors \[check box\]](#)

_3D.Width [SimTalk]

Sets the width of the *length-oriented object* designated by <Path>.

Type

Attribute

Syntax

```
<Path>._3D.Width:real
```

Assignment Value

You can assign a value of data type *real*.

Example

```
Conveyor._3D.Width := 2
```

See also

[Width \[tab Appearance\]](#) in the dialog **Edit 3D Properties**

Accessing Functions of the Extrusion Profile

_Accessing Functions of the Extrusion Profile

SimTalk provides the *functions* listed in the table of contents to the left for accessing an extrusion profile.

F3DgenerateEllipse [SimTalk]

Creates a two-dimensional ellipse profile which you can use in your user-defined extrusion configuration.

Type

Method

Syntax

```
F3DgenerateEllipse(BasePoint:length[2], Width:length, Height:length,
CurvePrecision:length)
```

Parameters

You can specify the following parameters:

- The parameter *BasePoint* is an *array* of data type *length* with two values which set the base point of the ellipse profile.
- The parameter *Width* of data type *length* sets its width.
- The parameter *Height* of data type *length* sets its height.
- The parameter *CurvePrecision* of data type *length* sets the sampling accuracy for the generate ellipse.

Example

```
var profile : any := F3DgenerateEllipse([2.6,4.5], 4, 4, 0.1)
// returns a two-dimensional array containing the values defining a profile
// to be used for an extrusion profile
```

See also

[_3D.ExtConfiguration \[SimTalk\]](#)

[_3D.InheritGraphics \[SimTalk\]](#)

[createExtrusionGraphic \[SimTalk\]](#)

[F3DgenerateHollowEllipse \[SimTalk\]](#)

[F3DgenerateHollowRectangle \[SimTalk\]](#)

[F3DgenerateRectangle \[SimTalk\]](#)

F3DgenerateHollowEllipse [SimTalk]

Creates a two-dimensional hollow ellipse profile which you can use in your user-defined extrusion configuration.

Type

Method

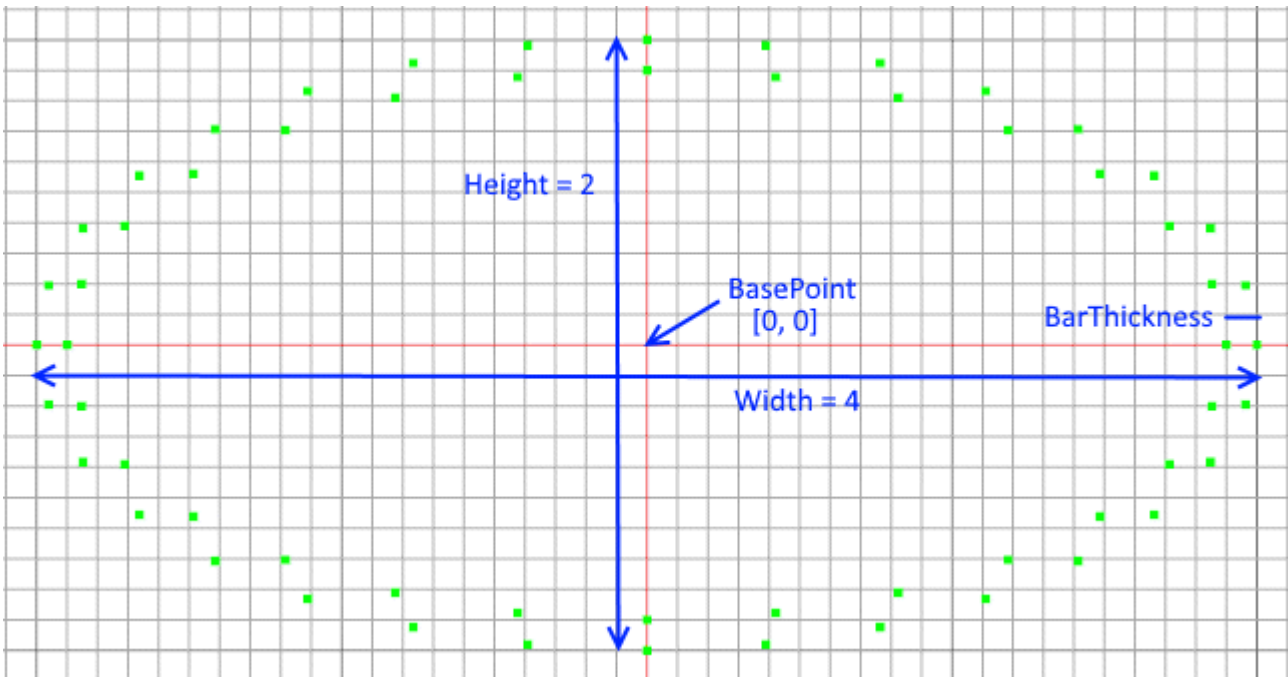
Syntax

```
F3DgenerateHollowEllipse(BasePoint:length[2], Width:length, Height:length,  
BarThickness:length, CurvePrecision:length)
```

Parameters

You can specify the following parameters:

- The parameter *BasePoint* is an *array* of data type *length* with two values which set the base point of the hollow ellipse profile.
- The parameter *Width* of data type *length* sets its width.
- The parameter *Height* of data type *length* sets its height.
- The parameter *BarThickness* of data type *length* sets the thickness of the bar.
- The parameter *CurvePrecision* of data type *length* sets the sampling accuracy for the generate ellipse.



Example

```
// The following code visualizes the generated profile using cubes
var points := F3DgenerateHollowEllipse([0,0], 4, 2, 0.1, 0.1)

for var i := 1 to points.ydim
    var Cube := _3D.getGraphic("deco").createCuboid([0.03, 0.03, 0.01])
    Cube.Position := [points[1,i], points[2,i], 0.0]
    Cube.MaterialActive := true
    Cube.MaterialDiffuseColor := makeRGBValue(0,255,0)
next
```

See also

[_3D.InheritGraphics \[SimTalk\]](#)

[_3D.ExtConfiguration \[SimTalk\]](#)

[createExtrusionGraphic \[SimTalk\]](#)

[F3DgenerateEllipse \[SimTalk\]](#)

[F3DgenerateHollowRectangle \[SimTalk\]](#)

[F3DgenerateRectangle \[SimTalk\]](#)

F3DgenerateHollowRectangle [SimTalk]

Creates a two-dimensional hollow rectangular profile which you can use in your user-defined extrusion configuration.

Type

Method

Syntax

```
F3DgenerateHollowRectangle(BasePoint:length[2], Width:length,  
Height:length, BarThickness:length)
```

Parameters

You can specify the following parameters:

- The parameter *BasePoint* is an *array* of data type *length* with two values which set the base point of the hollow rectangular profile.
- The parameter *Width* of data type *length* sets its width.
- The parameter *Height* of data type *length* sets its height.
- The parameter *BarThickness* of data type *length* sets the thickness of the bar.

Example

```
var profile : any := F3DgenerateHollowRectangle([2.6,4.5], 4, 2, 3, 0.1)  
// returns a two-dimensional array containing the values defining a profile  
// to be used for an extrusion profile
```

See also

[_3D.ExtConfiguration \[SimTalk\]](#)

[_3D.InheritGraphics \[SimTalk\]](#)

[createExtrusionGraphic \[SimTalk\]](#)

[F3DgenerateEllipse \[SimTalk\]](#)

[F3DgenerateHollowEllipse \[SimTalk\]](#)

[F3DgenerateRectangle \[SimTalk\]](#)

F3DgenerateRectangle [SimTalk]

Creates a two-dimensional rectangular profile which you can use in your user-defined extrusion configuration.

Type

Method

Syntax

```
F3DgenerateRectangle(BasePoint:length[2], Width:length, Height:length)
```

Parameters

You can specify the following parameters:

- The parameter *BasePoint* is an *array* of data type *length* with two values which set the base point of the rectangular profile.
- The parameter *Width* of data type *length* sets its width.
- The parameter *Height* of data type *length* sets its height.

Example

```
var profile : any := F3DgenerateRectangle([2.6,4.5],4, 2)  
// returns a two-dimensional array containing the values defining a profile  
// to be used for an extrusion profile
```

See also

[_3D.InheritGraphics \[SimTalk\]](#)

[createExtrusionGraphic \[SimTalk\]](#)

[F3DgenerateEllipse \[SimTalk\]](#)

[F3DgenerateHollowEllipse \[SimTalk\]](#)

[F3DgenerateHollowRectangle \[SimTalk\]](#)

[_3D.ExtConfiguration \[SimTalk\]](#)